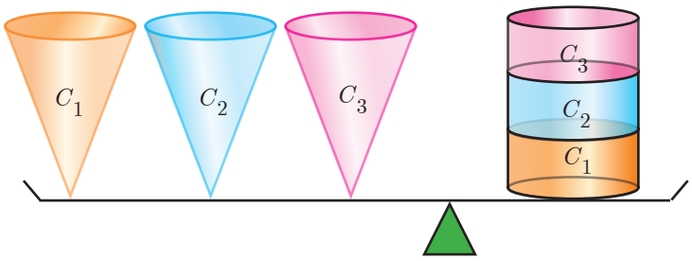
7.3.3 Volume of a right circular cone

Let r and h be the radius and height of a cone then its volume

$$V = \frac{1}{3}\pi r^2 h \text{ cu. units.}$$

Demonstration



- Consider a right circular cylinder and three right circular cones of same base radius and height as that of the cylinder.
- The contents of three cones will exactly occupy the cylinder.

Fig. 7.28

From, Fig.7.28 we see that,

$$3\times \text{(Volume of a cone)} = \text{Volume of cylinder}$$

$$= \pi r^2 h \text{ cu. units}$$

$$\text{Volume of a cone} = \frac{1}{3}\pi r^2 h \text{ cu. units}$$

Example 7.19 The volume of a solid right circular cone is 11088 cm^3. If its height is 24 cm then find the radius of the cone.

Solution Let r and h be the radius and height of the cone respectively.

Given that, volume of the cone $= 11088$ cm^3

$$\frac{1}{3}\pi r^2 h = 11088$$

$$\frac{1}{3}\times\frac{22}{7}\times r^2 \times 24 = 11088$$

$$r^2 = 441$$

Therefore, radius of the cone $r = 21$ cm

Thinking Corner

1. Is it possible to find a right circular cone with equal
   (a) height and slant height (b) radius and slant height (c) height and radius.
2. There are two cones with equal volumes. What will be the ratio of their radius and height?

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